[INCH-POUND]
A-A-59180
March 30,1998
SUPERSEDING
MIL-L-23225C
April 22, 1985
MIL-L-80061B
August 3, 1983

COMMERCIAL ITEM DESCRIPTION

LATHE, BENCH AND JEWELER'S, HORIZONTAL SPINDLE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal-spindle, bench and jeweler's lathes used in operations that require small metal workpiece turning, drilling, facing, or precise secondary finishing.

2. CLASSIFICATION. The lathes shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I – Bench lathe
Type II – Jeweler's lathe

Size – (See tables I and II)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC3416

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TABLE I. Type I machine sizes.

Size	Swing over bed, inches	Center-to-center distance, inches
0308	3	8
0317	3	17
0511	5	11
0710	7	10
0715	7	15
0819	8	19
0833	8	33
1118	11	18
1120	11	20
1125	11	25
1224	12	24
1236	12	36
1334	13	34
1336	13	36
1340	13	40

Note: The sizes shown are minimum requirements.

TABLE II. Type II machine sizes.

Size	Swing over bed, inches	Center-to-center distance, inches
0304	3	4
0607	6	7
0608	6	8
0611	6	11
0612	6	12
0807	8	7
0813	8	13
0815	8	15
0817	8	17

Note: The sizes shown are minimum requirements.

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The lathes shall include, as a minimum, the following components:

3.2.1 Bed. The bed shall support all the components of the bench lathe and shall have provisions for mounting to a benchtop or cabinet.

3.2.2 Ways. The ways shall be hardened and precision ground.

3.2.3 Spindle. The spindle shall be capable of multiple speeds through belt or gear drives, as specified (see 7.2(c)). The spindle shall have sealed and permanently lubricated bearings. If required, the spindle shall be equipped with the following, as specified (see 7.2(d)):

a. Foot-operated, hand-operated, or automatic brake.
b. Reverse rotation capability.
c. Powered collet closer.

3.2.4 Material holder. The lathe shall have material-holding mechanisms such as draw-in collets, multiple-jaw chucks, or shank-holding Morse tapers, as specified (see 7.2(e)). When specified (see 7.2(f)), the spindle shall accommodate all three methods of material holding.

3.2.5 Tailstock. The lathe shall have a tailstock that is adjustable and lockable along the ways. The tailstock shall accommodate material-holding centers and Morse taper tool inserts for drilling and boring operations.

3.2.6 Type I lathes.

3.2.6.1 Tool holder. The tool holder on the type I lathe shall be mounted on a carriage and cross-slide. The carriage shall hold the cross-slide and the tool post. The carriage shall move along the workpiece axis on the ways by hand crank or power feed, as specified (see 7.2(g)). Unless otherwise specified (see 7.2(h)), the power-feed lead screw shall be capable of being coupled to the spindle to coordinate feed and rotation for cutting threads. The cross-slide shall provide tool movement along the axis perpendicular to the workpiece axis and through the swing of the lathe. The cross-slide shall be equipped with a compound rest that shall allow 360-degree adjustment of the tool with respect to the workpiece axis. When specified (see 7.2(i)), the lathe shall be provided with a multiple tool-holding turret that can be installed on the ways in place of the tailstock. The turret shall possess a minimum of four tool positions, and shall be indexable with independently adjustable stops for each turret position. The turret shall have a means to feed and retract the indexed tool into the workpiece.

3.2.6.2 Controls. Type I lathe controls shall, as a minimum, consist of spindle speed, tool feed, cutting depth, and start/stop. Unless otherwise specified (see 7.2(j)), control adjustments shall be accomplished by hand wheels, locking levers, or control knobs. Type I lathes with a powered tool

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feed leadscrew shall have a means to adjust tool feed rate and thread cutting pitch.

3.2.7 Type II lathes.

3.2.7.1 Tool holder. As a minimum, type II lathes shall be equipped with slides that allow tool movement along the workpiece axis and perpendicular to the workpiece axis to allow turning, facing, and boring. When specified (see 7.2(k)), the slides shall be powered. When specified (see 7.2(l)), the lathe shall be provided with a multiple tool-holding turret that can be installed on the ways. The turret shall possess a minimum of four tool positions, and shall be indexable with independently adjustable stops for each turret position. The turret shall have a means to feed and retract the indexed tool into the workpiece.

3.2.7.2 Controls. The type II lathe controls shall, as a minimum, consist of spindle speed, tool feed, cutting depth, and start/stop. Unless otherwise specified (see 7.2(m)), control adjustments shall be accomplished by hand wheels, locking levers, or control knobs. When specified (see 7.2(n)), controls shall be provided that allow tool depth and feed times to be preset and then controlled automatically upon manual start. When specified (see 7.2(o)), the bench lathe shall provide automatic stop and workpiece ejection at the end of the turning operation.

3.2.8 Lubrication system. Unless otherwise specified (see 7.2(p)), machine components requiring continuous lubrication shall be permanently lubricated or lubricated by automatic means. Machine components requiring periodic manual lubrication shall be readily accessible.

3.2.9 Coolant system. When specified (see 7.2(q)), the bench lathe shall be equipped with a tool coolant system. The coolant system shall consist of a reservoir or sump, powered pump, chip tray, and nozzle(s).

3.2.10 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79 or International Electrotechnical Commission (IEC) 204-1. Unless otherwise specified (see 7.2(r)), the bench lathe shall operate from a dual 230/460-volt, single-phase, 60-hertz circuit, and shall be wired for 230-volt operation. The control system shall operate on a 120-volt, single-phase, 60-hertz circuit.

3.2.11 Electric motors. The spindle drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(s)), the motors shall have drip-proof enclosures.

3.2.12 Optional components. Optional components shall be provided as specified (see 7.2(t)).

3.3 Performance. Unless otherwise specified (see 7.2(u)), the lathes shall meet the performance requirements of table III when tested in accordance with sections 5.3 through 5.3.4 herein.

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TABLE III. Performance requirements.

Parameter	Type I	Type II
Collet run-out, inch	0.0002	0.0002
Spindle run-out, inch	0.0005	0.0005
Cylindrical turn:		
Roundness, inch	0.001	—
Taper, inch	0.002	—
Facing flatness, inch	—	0.001

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(v)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.6. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(w)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(x)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(y)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer’s name.
c. Machine model designation.
d. Power input (volts, total amperes, phase, frequency).
e. Short-circuit/over-current rating.
f. Contract number or order number.
g. National stock number or plant equipment code.
h. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(z)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.1 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same system offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The bench lathe shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(aa)), the bench lathe shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(bb)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct the tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(cc)). Unless otherwise specified (see 7.2(dd)), the machine shall be tested under environmental conditions deemed appropriate by the manufacturer.

5.3.2 Operational test. The lathe shall be operated in accordance with the manufacturer's standard operating procedures for a period of 1 hour. Proper operation of all controls and features shall be verified during the test period.

5.3.3 Performance tests. Unless otherwise specified (see 7.2(ee)), the performance test shall consist of spindle run-out, cylindrical turning, and facing tests.

5.3.3.1 Spindle run-out. The spindle run-out test shall consist of obtaining the measurement data for the parameters for spindle and collet run-out as specified in test numbers 4 and 20 of ANSI B5.16. The measurements shall not exceed the performance requirements shown in table III for the particular test. The measuring instrument(s) shall be the manufacturer's choice.

5.3.3.2 Cylindrical turning test. Type I lathes shall undergo the cylindrical turning test of ANSI B5.16, test numbers 21 and 23. The test workpiece shall be American Iron and Steel Institute (AISI) 1020 steel. The workpiece length and diameter shall be 1/2 the maximum swing and

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center-to-center capacity of the lathe. The tool choice and depth of test cut shall be at the manufacturer's discretion. The length of the test cut shall be 3/4 of the length of the test workpiece. The roundness and taper over the test cut length measurements shall meet the requirements shown in table III.

5.3.3.3 Facing test. Type II lathes shall undergo a facing test. The test workpiece shall be AISI 1020 steel. The workpiece length and diameter shall be 1/2 the maximum capacity of the lathe. The tool choice and depth of cut shall be at the manufacturer's discretion. Facing flatness shall not exceed the value shown in table III.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(ff)).

5.4 Acceptance. Unless otherwise specified (see 7.2(gg)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance requirements shown in table III. Failure of the system to meet the performance requirements shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (7.2(hh)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.6	Machine Tools – Lathes – Safety Requirements for Construction, Care, and Use
ANSI B5.16	Accuracy of Engine and Tool Room Lathes – One of a Series of Standards for Small Tools and Machine Tool Elements

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

International Electrotechnical Commission (IEC)

IEC 204-1	Electrical Equipment of Industrial Machines - Part I: General Requirements

Applications for copies should be sent to: American National Standards Institute, 11 West

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42nd Street, New York, NY 10036.

National Electrical Manufacturers Association (NEMA)

NEMA/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to: National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

Occupational Safety and Health Association (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR Part 1910)

Applications for copies should be sent to: U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size of bench lathe required (see 2.1).
c. Specify spindle speeds and belt or gear drive, if required (see 3.2.3).
d. Spindle brake, reverse rotation, power collet closer, if required (see 3.2.3).
e. Specify draw-in collet, multiple-jaw chuck, or taper, if required (see 3.2.4).
f. Specify all three material holding methods, if required (see 3.2.4).
g. Hand crank or power feed, as specified (see 3.2.6.1).
h. Lead screw, if different (see 3.2.6.1).
i. Tool-holding turret, if required (see 3.2.6.1).
j. Type I control adjustments, if different (see 3.2.6.2).
k. Powered slides, if required (see 3.2.7.1).
l. Tool-holding turret for type II lathes, if required (see 3.2.7.1).
m. Type II control adjustments, if different (see 3.2.7.2).
n. Automatic tool depth and feed times, if required (see 3.2.7.2).
o. Automatic stop and workpiece ejection, if required (see 3.2.7.2).
p. Lubrication system, if different (see 3.2.8).
q. Coolant system, if required (see 3.2.9).
r. Electrical system, if different (see 3.2.10).
s. Motor enclosure, if different (see 3.2.11).
t. Optional components, as specified (see 3.2.12).
u. Performance, if different (see 3.3).
v. Machine dimensions and weight, if required (see 3.4).
w. System of units required (see 3.6).
x. Dual system of units, if required (see 3.6).
y. Nameplate, if different (see 3.7).
z. Lubrication plate or chart, if required (see 3.8).
aa. Testing, if different (see 5.3).

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bb. Test responsibility, if different (see 5.3.1).
cc. Calibration certification, if required (see 5.3.1).
dd. Environmental conditions, if different (see 5.3.1).
ee. Performance tests, if different (see 5.3.3).
ff. Optional tests, as specified (see 5.3.4).
gg. Acceptance, if different (see 5.4).
hh. Packaging requirements (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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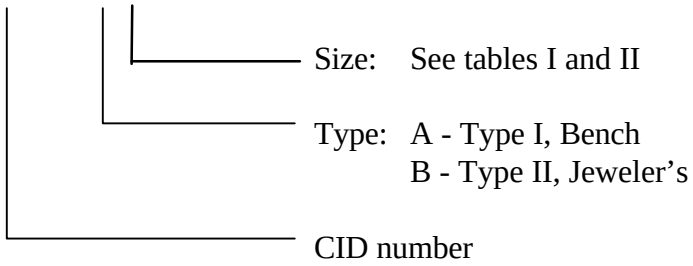

MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Army - AV
Navy - MC

PREPARING ACTIVITY
DLA - GS

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